Setup Guide OpenVPN Master branch
Version 0.3

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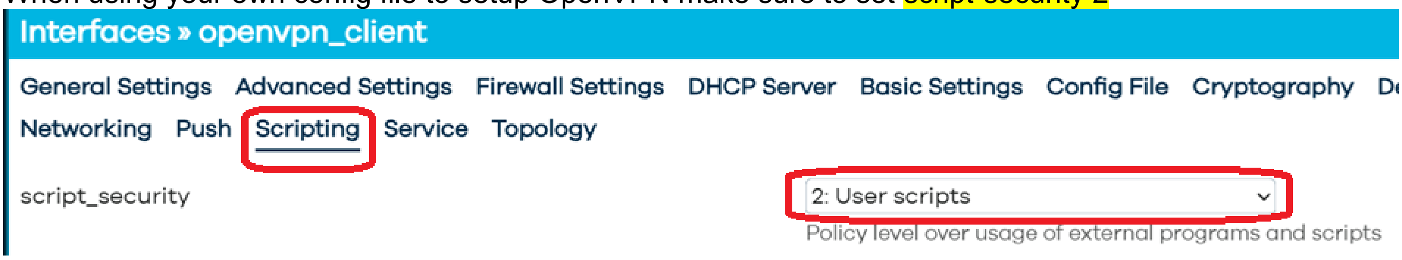
Introduction
On master branch *luci-app-openvpn* is replaced by *luci-proto-openvpn*.
There is no longer a separate entry in the GUI for OpenVPN but it has become a protocol so it is integrated in the interface.

This guide and also the OpenVPN implementation is very much a Work in Progress and subject to change.

When using your own config file to setup OpenVPN make sure to set script-security 2



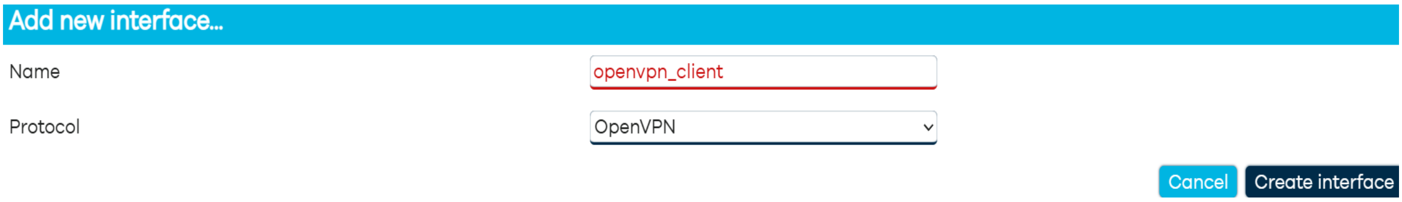
This will allow you to setup the VPN tunnel to you own liking

/var/run/openvpn.openvpn_client.conf:

```
cd /etc/openvpn/openvpn_client
status /var/run/openvpn.openvpn_client.status
syslog openvpn_openvpn_client
tmp-dir /tmp
script-security 2
dev tun1
dev-type tun
mtu-disc no
config '/etc/openvpn/openvpn_client/openvpn_client_config.cfg'
```

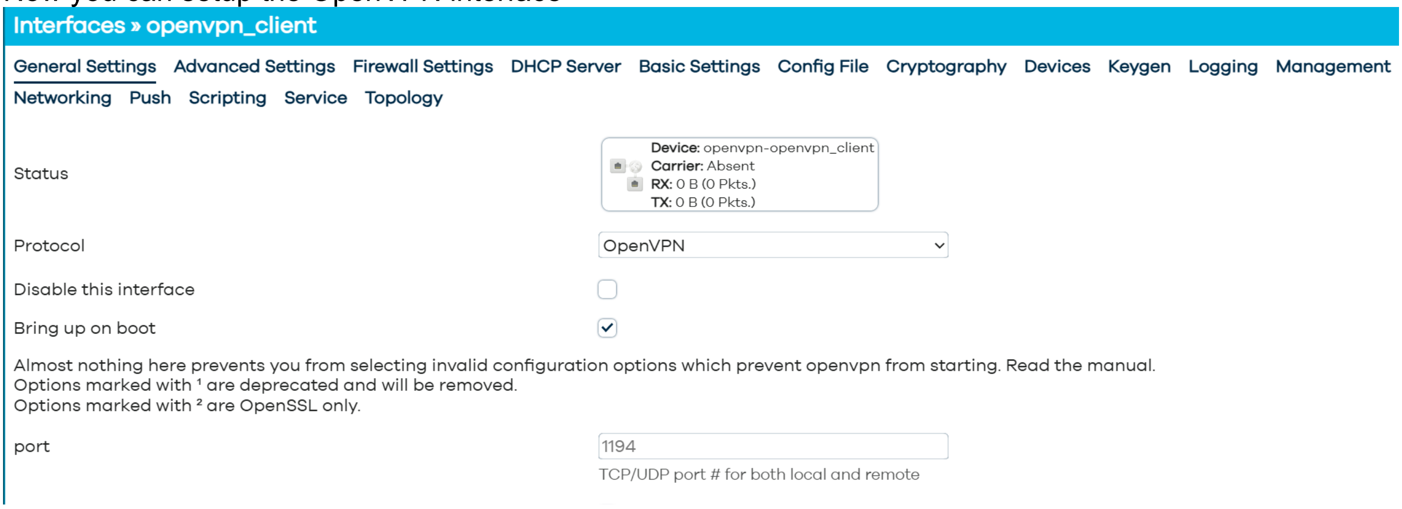
Setup Interface

To setup OpenVPN (whether client or server)
Network > Interfaces > Add new interface
Name: choose a name not more than 15 characters, use *openvpn_client*
Protocol: OpenVPN.



Click:
Create interface

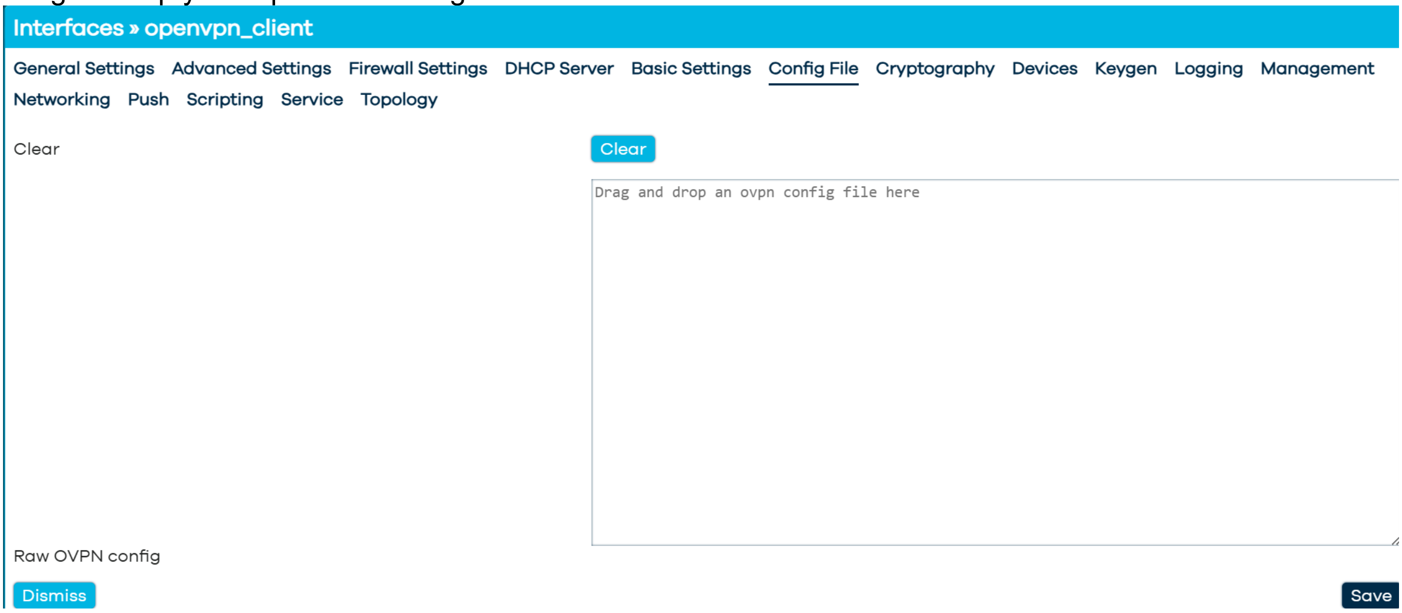
Now you can setup the OpenVPN interface



There are two ways to setup.

The easiest way is to import your config file into the interface.
Click: Config File

Drag en drop your OpenVPN config file in the box



Config files can contain all necessary info *inline* including keys certificates and username and password as shown below

Interfaces » openvpn_client

General Settings Advanced Settings Firewall Settings DHCP Server Basic Settings Config File
Networking Push Scripting Service Topology

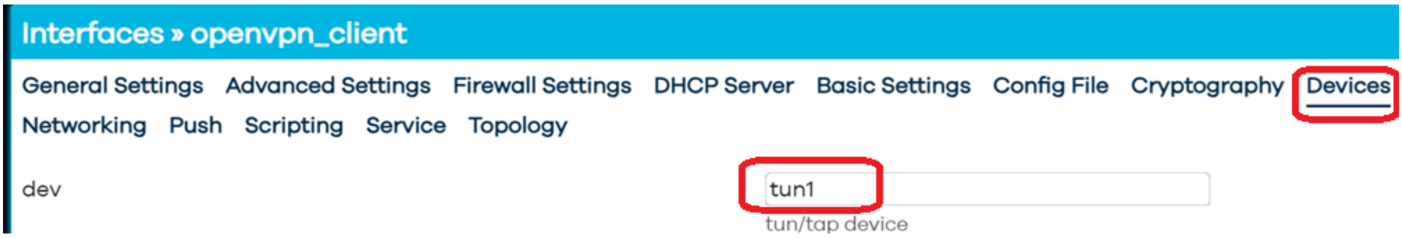
Clear Clear

Raw OVPN config

```
remote fr-rbx.vpnunlimitedapp.com
proto udp
port 1197

<auth-user-pass>
REDACTED
</auth-user-pass>

client
dev tun1
reneg-sec 0
persist-tun
ping 5
nobind
allow-compression no
remote-cert-tls server
auth-nocache
data-ciphers CHACHA20-POLY1305:AES-256-GCM:AES-128-GCM
auth sha512

pull-filter ignore "redirect-gateway def1"

#for hotplug scripts use 3:
script-security 3

<ca>
-----BEGIN CERTIFICATE-----
redacted
-----END CERTIFICATE-----
-----BEGIN CERTIFICATE-----
redacted
-----END CERTIFICATE-----
</ca>

<cert>
-----BEGIN CERTIFICATE-----
redacted
-----END CERTIFICATE-----
</cert>

<key>
-----BEGIN PRIVATE KEY-----
redacted
-----END PRIVATE KEY-----
</key>
```

Often a config file just has *dev tun* so without a number, in this case OpenVPN itself will use the first available number which is **0** so the device will be tun0
I think this is bad practice and encourage you to choose a number for the device.
I use *tun1* for my client and *tun2* for my server.

As is chosen to setup the tunnel with a config file with script-security 2 there is no information about the used *l3_device* (the tun1 device in this example) to make sure applications which use the interface can extract the l3_device from the config (e.g. for Policy Based Routing) also add the device under Devices > dev:



this way PBR wil now the device used by the openvpn_client interface

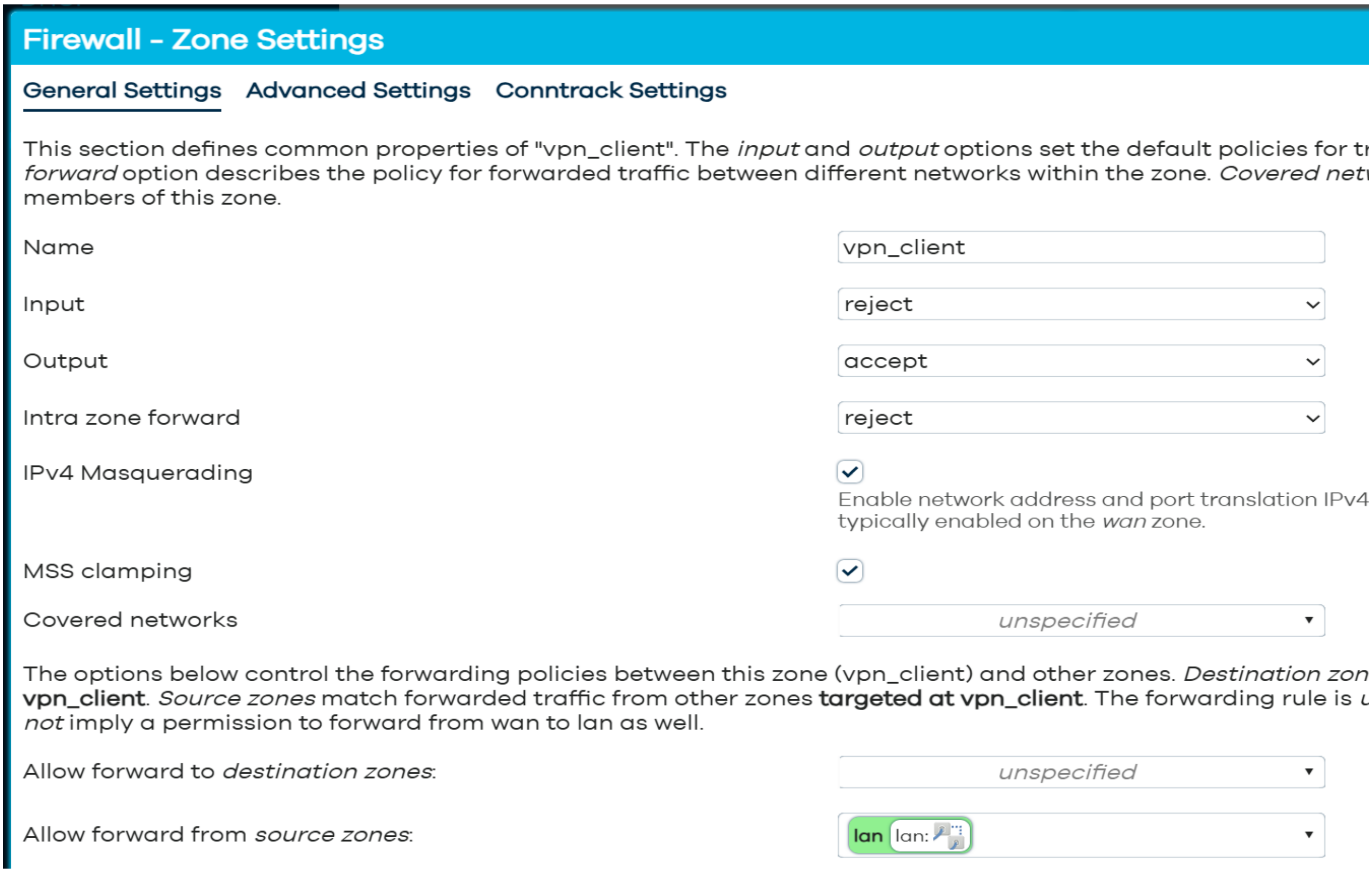
/etc/config/network/

```
config interface 'openvpn_client'
        option proto 'openvpn'
        option dev_type 'tun'
        option mtu_disc 'no'
        option multipath 'off'
        option config '/etc/openvpn/openvpn_client/openvpn_client_config.cfg'
        option script_security '2'
        option dev 'tun1'
```

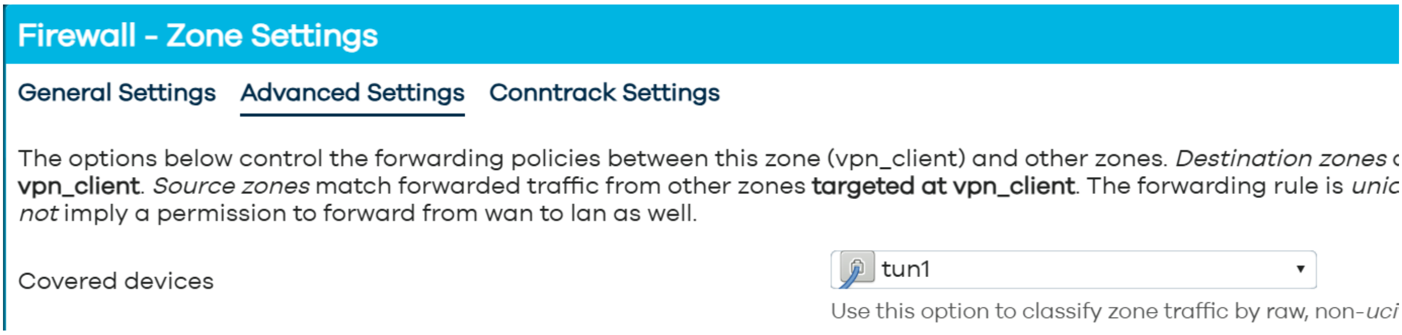
Setup Firewall

OpenVPN Client

Create a new firewall zone:
Network > Firewall > Add
Name: vpn_client
Input: **reject**
Output: **accept**
Intra zone forward: **reject**
IPv4 Masquerading: **Enabled** (ticked)
MSS clamping: **Enabled** (ticked)
Covered Networks: none
Allow forward from source zones: **lan**



In this example I have adapted my config file to use: dev tun1
As we have used script-security 2 and use our own config there is no device information available for the firewall to use so we have to add the device on the Advanced Settings tab under Covered devices:



Alternatively you can add an extra interface with name: *tun1_client*, protocol: *unmanaged* and device: *tun1*
You can the use this interface *tun1_client* to add under Covered networks instead of adding the device under Covered devices

OpenVPN Server

Open up firewall for the server port

Allow Input, Output and Intra zone forward

Allow from VPN to LAN , from VPN to WAN if you want internet for the connected clients and Allow from LAN to VPN if you want bidirectional traffic